Rational Functions and Asymptotes

Section 1: Vertical Asymptotes and Holes

1 Find all vertical asymptotes and holes for $f(x) = \frac{x^2 - 4}{x^2 - x - 2}$.

a. Hole at $x = 2$, Vertical asymptote at $x = -1$

b. Vertical asymptotes at $x = 2$ and $x = -1$

c. Hole at $x = -1$, Vertical asymptote at $x = 2$

d. No holes, vertical asymptote at $x = 2$

2 Find the coordinates (x, y) of the hole for $f(x) = \frac{x^2 - 9}{x - 3}$.

a. $(3, 6)$

b. $(3, 0)$

c. $(3, 3)$

d. $(-3, 0)$

3 How do you determine if a real zero of the denominator causes a hole or a vertical asymptote?

a. If the numerator also has the same zero of equal or greater multiplicity, it is a hole; otherwise it is a vertical asymptote

b. Every zero of the denominator is an asymptote

c. If the degree of the numerator is larger, it is always a hole

d. Check the sign of $f(0)$

Section 2: Horizontal Asymptotes

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4 Find the horizontal asymptote of $f(x) = \frac{4x^2 + 1}{3x^2 - 5x + 2}$.

a. $y = 4/3$

b. $y = 0$

c. No horizontal asymptote

d. $y = 1/2$

5 Find the horizontal asymptote of $g(x) = \frac{7x - 2}{x^3 + 4}$.

a. $y = 0$

b. $y = 7$

c. No horizontal asymptote

d. $y = -1/2$

6 Under what condition does a rational function have NO horizontal asymptote?

a. When the degree of the numerator is strictly greater than the degree of the denominator

b. When the degree of the denominator is greater

c. When the degrees are equal

d. When there are no real roots

Section 3: Slant (Oblique) Asymptotes

7 When does a rational function $f(x) = \frac{P(x)}{Q(x)}$ possess a slant (oblique) asymptote?

a. When the degree of the numerator is exactly 1 greater than the degree of the denominator

b. When the degree of the numerator is 2 greater

c. When the degree of the denominator is greater

d. When there are no vertical asymptotes

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8 Find the slant asymptote of $f(x) = \frac{2x^2 + 7x + 5}{x + 2}$.

a. $y = 2x + 3$

b. $y = 2x + 7$

c. $y = 2x - 1$

d. $y = x + 3$

9 Find the slant asymptote of $g(x) = \frac{x^3 - 1}{x^2 + 1}$.

a. $y = x$

b. $y = x - 1$

c. $y = 1$

d. No slant asymptote

Section 4: Intercepts and Asymptote Crossings

10 Find all x-intercepts and the y-intercept of $f(x) = \frac{x^2 - 1}{x^2 - 4}$.

a. x-intercepts: $(\pm 1, 0)$, y-intercept: $(0, 1/4)$

b. x-intercepts: $(\pm 2, 0)$, y-intercept: $(0, -1/4)$

c. x-intercepts: $(\pm 1, 0)$, y-intercept: $(0, -1)$

d. x-intercepts: $(1, 0)$, y-intercept: $(0, 1/4)$

11 Can a rational function cross its horizontal or slant asymptote?

a. Yes, a function can cross its horizontal or slant asymptote at finite values of x

b. No, a graph can never touch any asymptote

c. Only at the y-intercept

d. Only if it has no vertical asymptotes

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12 Find the point where $f(x) = \frac{x^2 + 3x - 4}{x^2 - 1}$ crosses its horizontal asymptote $y = 1$.

a. $(-1, 1)$ is undefined; crossing is at $x = 1$ is undefined; crosses at $(1, 1)$ is a hole; no crossing point

b. It crosses at $x = 1$

c. It crosses at $x = 0$

d. It does not cross its horizontal asymptote

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